

Chapter 5: Permutation Groups

1. Write the following permutations in cycle notation.

$$(b) \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 4 & 2 & 5 & 1 & 3 \end{pmatrix} = (14)(35)$$

$$(d) \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 4 & 3 & 2 & 5 \end{pmatrix} = (42)$$

2. Compute each of the following.

$$(d) (1423)(34)(56)(1324) = (12)(56)$$

$$(i) \text{ Here we will use the facts that } (1254)^{-2} = ((1254)^{-1})^2 \text{ and } (1254)^{-1} = (4521).$$

$$\begin{aligned} (123)(45)(1254)^{-2} &= (123)(45)((1254)^{-1})^2 \\ &= (123)(45)(4521)^2 \\ &= (123)(45)(4521)(4521) \\ &= (143)(25) \end{aligned}$$

$$(j) (1254)^{100} = ((1254)^4)^{25} = (1)^{25} = (1)$$

$$(k) |(1254)| = 4$$

3. Write the following permutations as products of transpositions and identify them as even or odd.

$$(b) (156)(234) = (15)(56)(23)(42) \dots \dots \dots \text{This permutation is even.}$$

$$(d) (142637) = (13)(12)(37)(26)(14) \dots \dots \dots \text{This permutation is odd.}$$

10. Find an element of largest order in S_n for $n = 3, 4, 5, 6, 7, 8, 9, 10$.

Recall that every permutation is a product of disjoint cycles. Also, the order of a product of disjoint cycles is the least common multiple of the orders of the cycles. For example the permutation $(123)(4567)(89)$ has order $\text{lcm}(3, 4, 2) = 12$. Thus we can find an element of largest order in S_n by looking for a product of disjoint cycles, such that the lcm of the orders of the cycles is as large as possible.

$$(a) \text{ An element of largest order in } S_3 \text{ is } (123), \text{ and it has order } 3.$$

$$(b) \text{ An element of largest order in } S_4 \text{ is } (1234), \text{ and it has order } 4.$$

$$(c) \text{ An element of largest order in } S_5 \text{ is } (12)(345), \text{ and it has order } 2 \cdot 3 = 6.$$

$$(d) \text{ An element of largest order in } S_6 \text{ is } (123456), \text{ and it has order } 6.$$

$$(e) \text{ An element of largest order in } S_7 \text{ is } (123)(4567), \text{ and it has order } 3 \cdot 4 = 12.$$

$$(f) \text{ An element of largest order in } S_8 \text{ is } (123)(45678), \text{ and it has order } 3 \cdot 5 = 15.$$

$$(g) \text{ An element of largest order in } S_9 \text{ is } (1234)(56789), \text{ and it has order } 4 \cdot 5 = 20.$$

$$(h) \text{ An element of largest order in } S_{10} \text{ is } (01)(234)(56789), \text{ and it has order } 2 \cdot 3 \cdot 5 = 30.$$

23. If σ is a cycle of odd length, then σ^2 is also a cycle.

Proof. Suppose $\sigma \in S_n$ is a cycle of odd length $2k + 1$. Then we have

$$\sigma = (a_0 \ a_1 \ a_2 \ a_3 \ a_4 \ \dots \ a_{2k-1} \ a_{2k}),$$

for some subset $X = \{a_0, a_1, a_2, \dots, a_{2k}\} \subseteq \{1, 2, 3, 4, \dots, n\}$. This means that

$$\sigma(a_i) = a_{i+1}$$

for each $a_i \in X$ (where the $i + 1$ is computed mod $2k + 1$), while $\sigma(x) = x$ for each $x \in \{1, 2, 3, \dots, n\} \setminus X$.

Now consider the effect of $\sigma^2 = \sigma\sigma$. The above paragraph implies that for each $a_i \in X$ we have $\sigma^2(a_i) = \sigma(\sigma(a_i)) = \sigma(a_{i+1}) = a_{i+2}$, where, again, the addition $i + 2$ is done modulo $2k + 1$ and $\sigma^2(x) = x$ for any x not in X . From this we see that

$$\sigma^2 = (a_0 \ a_2 \ a_4 \ a_6 \ \dots \ a_{2k}, \ a_1 \ a_3 \ a_5, \ \dots \ a_{2k-1}),$$

and is therefore a cycle. ■

31. For $\alpha, \beta \in S_n$, define $\alpha \sim \beta$ if there is some $\sigma \in S_n$ for which $\sigma\alpha\sigma^{-1} = \beta$. Show that $\sim$ is an equivalence relation.

Proof. We need to show that $\sim$ is reflexive, symmetric and transitive.

- (1) Observe that for any $\alpha \in S_n$ we have $\sigma\alpha\sigma^{-1} = \alpha$, where $\sigma = \text{id} \in S_n$. Thus $\sim$ is reflexive.
- (2) Suppose $\alpha \sim \beta$. By definition of $\sim$ this means that there is some $\sigma \in S_n$ for which $\sigma\alpha\sigma^{-1} = \beta$. Left-multiplying both sides of this by σ^{-1} gives us $\alpha\sigma^{-1} = \sigma^{-1}\beta$. Now right-multiplying both sides by σ gives $\alpha = \sigma^{-1}\beta\sigma$. Rewrite this as

$$\sigma^{-1}\beta(\sigma^{-1})^{-1} = \alpha. \tag{1}$$

As $\sigma \in S_n$, we also have $\sigma^{-1} \in S_n$, so Equation (1) and the definition of $\sim$ now yields $\beta \sim \alpha$.

We've shown $\alpha \sim \beta$ implies $\beta \sim \alpha$, so $\sim$ is symmetric.

- (3) Now we will show that $\sim$ is transitive. Suppose $\alpha \sim \beta$ and $\beta \sim \gamma$. Then there are permutations $\sigma, \tau \in S_n$ for which $\sigma\alpha\sigma^{-1} = \beta$ and $\tau\beta\tau^{-1} = \gamma$. Inserting the value for β given by the first equation into the second and using associativity yields

$$\begin{aligned} \tau(\sigma\alpha\sigma^{-1})\tau^{-1} &= \gamma \\ (\tau\sigma)\alpha(\sigma^{-1}\tau^{-1}) &= \gamma \\ (\tau\sigma)\alpha(\tau\sigma)^{-1} &= \gamma. \end{aligned}$$

As $\tau, \sigma \in S_n$, we also have $\tau\sigma \in S_n$. Hence the above equation shows $\alpha \sim \gamma$.

Having shown that $\alpha \sim \beta$ and $\beta \sim \gamma$ together imply $\alpha \sim \gamma$, we see that $\sim$ is transitive.

The above considerations show that $\sim$ is an equivalence relation. ■

34. Suppose α is an even permutation. Prove that α^{-1} is also even. Is the same true if α is odd?

Proof. Suppose α is even. This means there is an even integer p and transpositions $\tau_1, \tau_2, \dots, \tau_p$ for which $\alpha = \tau_1\tau_2\tau_3 \cdots \tau_{p-1}\tau_p$. Now the inverse of each transposition τ_i is of course $\tau_i^{-1} = \tau_i$. Therefore we have

$$\begin{aligned} \alpha^{-1} &= (\tau_1\tau_2\tau_3 \cdots \tau_{p-1}\tau_p)^{-1} \\ &= \tau_p^{-1}\tau_{p-1}^{-1} \cdots \tau_3^{-1}\tau_2^{-1}\tau_1^{-1} \\ &= \tau_p\tau_{p-1} \cdots \tau_3\tau_2\tau_1. \end{aligned}$$

This expresses α^{-1} as a product of an even number of transpositions, so α^{-1} is even.

By the same reasoning, if α is odd, its inverse α^{-1} will be odd too. Thus it is not true that if α is odd, then its inverse is even. ■